Through Zero and Beyond – Multiple Steps at a Time

Questions

1. Fill in the missing numbers in each sequence.

a)

–6 ___ –2 ___ 2 ___ 6

b)

___ –25 –20 ___ –10 ___ 0

c)

–21 ___ –15 ___ –9 –6

d)

___ –40 –30 ___ –10 0

2. Complete the missing numbers.

a)

–18 –12 –6 ___ 6 12

b)

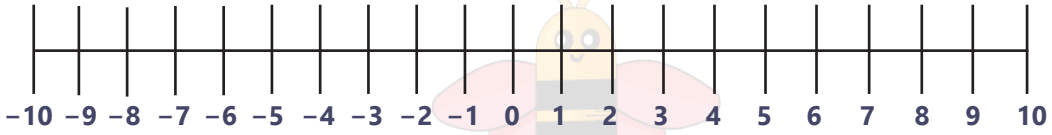
–16 –12 ___ –4 0 ___ 8

c)

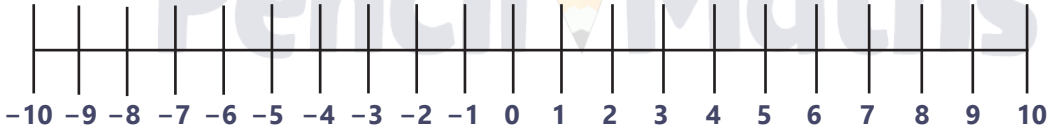
–9 –6 ___ 0 3 ___ 9

3. Continue the sequences using the number lines.

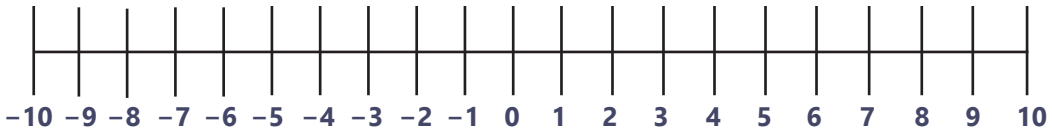
a) **6, 3, 0, _____, _____, _____**



b) **–10, –8, -6, _____, _____, _____**



c) **–9, –5, –1, _____, _____, _____**



4. Zak is counting backwards in **6s** starting from **0.**

What will the next number in Zak's sequence be after **–12**?

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5. Maya says, "If I start at –12 and count in 6s, I will never reach a positive number." Do you agree with Maya? Explain your reasoning and list the first three positive numbers she will reach (if any).

6. Arjun is counting forward in **4s** from **–8.** Sia is counting backward in **3s** from **6.** Write any one number that both **Arjun and Sia** will say.

7. Continue each sequence with the next three numbers.

a) 10, 6, 2 → _____, _____, _____

b) –10, –6, –2 → _____, _____, _____

c) –21, –17, –13 → _____, _____, _____

d) 15, 9, 3 → _____, _____, _____

8. Find the next three numbers in each pattern.

a) –12, –9, –6 → _____, _____, _____

b) –30, -25 , -20 → _____, _____, _____

c) –5, 0, 5 → _____, _____, _____

d) 30, 24, 18 → _____, _____, _____

9. A diver is at **–12 m** below the surface. He ascends **4 m** every minute. How many minutes will it take him to reach the surface (0 m)? Show your working:

10. Tanvi is counting back in 7s starting from 14. She says the next three numbers are: **14, 8, 2, –3**

a) What mistake has she made?

b) Write the correct next three numbers:

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